

Exam Title: “Turning Losses into Loyalty: Analyzing Churn Risk in Subscription-Based Services”

 **Exam Type:** Standalone Business Case-Based Practical

 **Duration:** 3 Hours

 **Dataset:** A gigantic CSV dataset: **CustomerSubscriptions.csv** (Generate from AI tool)

Project Task

You are a junior data analyst for a startup that offers subscription-based fitness and wellness services (e.g., yoga, gym, nutrition). The management team is concerned about increasing customer churn.

Instructions & Evaluation Criteria

 **Note:** You **must** use SQL for data analysis and then use Excel **OR** Power BI to visualize key metrics. Present your insights through charts and tables.

 **Dataset File:**

CustomerSubscriptions.csv

Column Name	Description
CustomerID	Unique customer ID

Name	Customer name
Age	Age of customer
Gender	Male / Female / Other
SubscriptionType	Monthly, Quarterly, Yearly
SubscriptionDate	When subscription started
LastLoginDate	Last time customer accessed the service
TotalSessions	Total sessions (yoga/gym/consultation) attended
FeedbackScore	Rating from 1 to 10
IsChurned	1 = Churned, 0 = Active

PART 1 – SQL Analysis

Perform the following queries using SQL:

-  Total number of active vs churned customers (by SubscriptionType).
-  Average FeedbackScore by SubscriptionType and Gender.
-  List customers who attended < 5 sessions AND gave feedback < 5.
-  Identify customers who haven't logged in for the past 60 days.
-  Churn rate by SubscriptionType.
-  List Top 10 customers with longest subscriptions (based on SubscriptionDate).
-  Age group-wise churn analysis (e.g., 18–25, 26–35, etc.).

Export results as CSV or temporary table for use in Excel/Power BI.

PART 2 – Excel or Power BI Dashboard

Use either the original dataset or your SQL outputs to build a dashboard that includes:

- Donut or Pie Chart** for Churned vs Active Customers
- Bar Chart:** Churn rate by SubscriptionType
- Line Chart:** Monthly churn trend (based on SubscriptionDate)

4. **Scatter Plot:** FeedbackScore vs TotalSessions (with color for Churned)

5. **KPIs:**

- Total Customers
- % Churned
- Avg Feedback Score
- Avg Days Since Last Login

Include slicers or filters for SubscriptionType, Gender, and Age Group.



Final Submission Checklist

File	Description
ChurnAnalysis.sql	SQL script for all queries
ChurnDashboard.xlsx or .pbix	Completed dashboard
ChurnSummary.txt	4–5 bullet points with insights and suggestions



Example Insights to Derive

- Are younger customers more likely to churn?
- Is monthly subscription less sticky than yearly?
- Is low session attendance a churn predictor?
- Are low-feedback customers more likely to leave?

Practical Exam BUSINESS CASE STUDY

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